5.9

Heat capacity of a solid a. we need to integrate this. it says to use gaussian quadrature so we'll try that but we'll also maybe try simpson's rule

In [2]:
```python
import numpy as np
import matplotlib.pyplot as plt
from gaussxw import gaussxwab
```

In [3]:
```python
# setting up constants
V = 0.001
rho = 6.022e28
kB = 1.38*10**(-23)
thD = 428
```

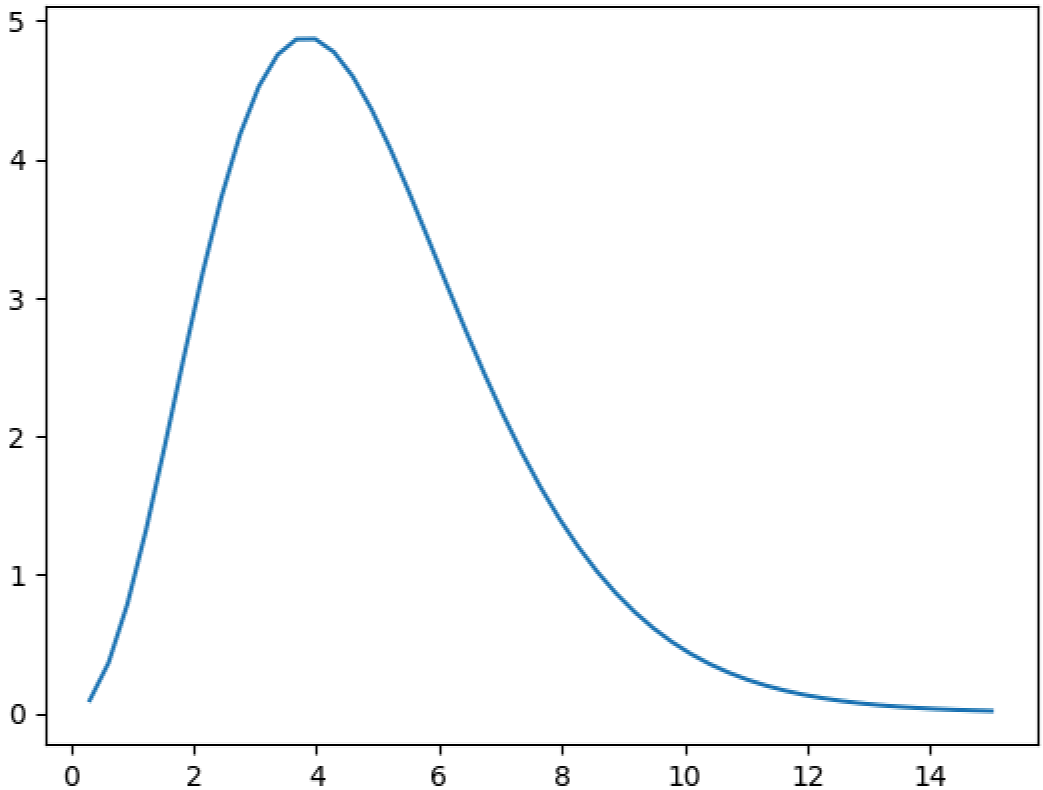
In [4]:
```python
def f(x):
    return(x**4*np.exp(x)/(np.exp(x)-1)**2)
```

Let's look at that function first just to get a sense for what it looks like.

In [6]:
```python
fig0, ax0 = plt.subplots()
x = np.linspace(0,15)
y = f(x)
ax0.plot(x, y)
```

```
/tmp/ipykernel_13601/1435524251.py:2: RuntimeWarning: invalid value encounte
red in divide
  return(x**4*np.exp(x)/(np.exp(x)-1)**2)
```

Out[6]:
```
[<matplotlib.lines.Line2D at 0x7997a59d7860>]
```



So that is the function that we want to integrate which is inside a function of T. So the function itself is an integral of part of the curve we see above.

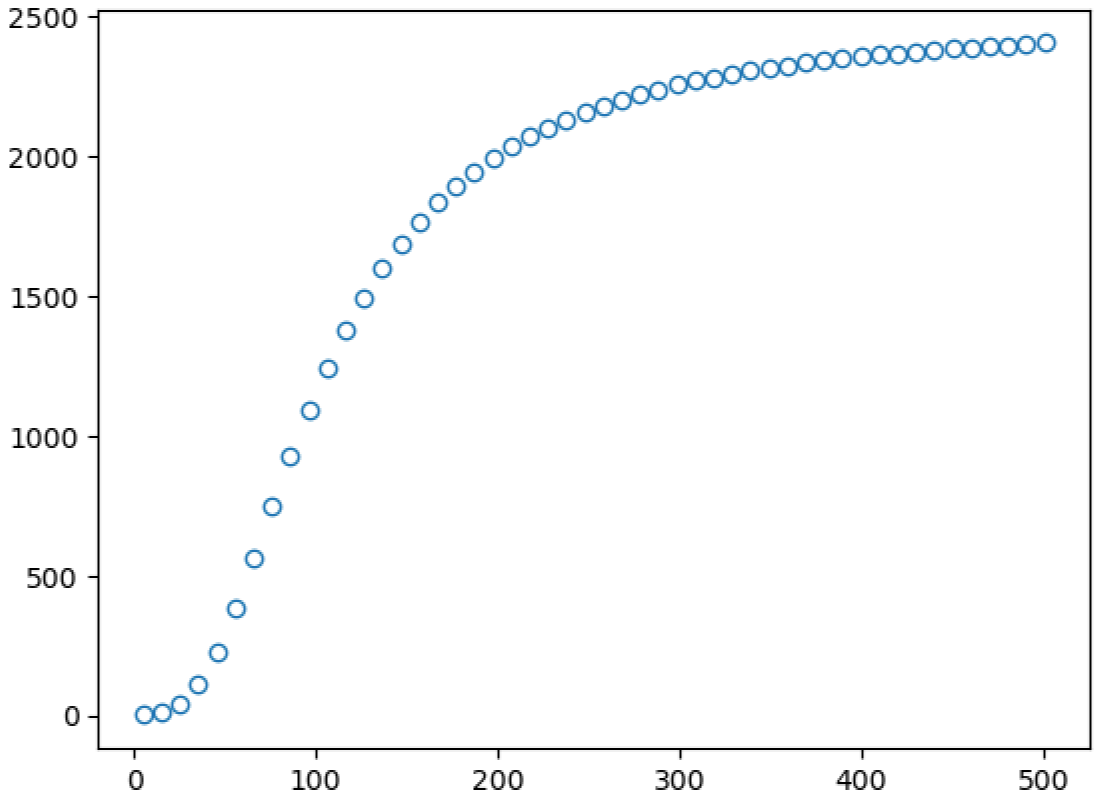
In [46]:
```
def cv(T):
    N = 50
    x, w = gaussxwab(N, 0, thD/T)
    s = 0.0
    for k in range(N):
        s = s + w[k]*f(x[k])
    return(9*V*rho*kB*((T/thD)**3)*s)
```

In [47]:
```
fig, ax = plt.subplots()

t = np.linspace(5, 501)
c = []
for i in t:
    c.append(cv(i))

ax.plot(t, c, 'o', mfc='none')
```

Out[47]:
```
[<matplotlib.lines.Line2D at 0x7a4011579cd0>]
```



In [ ]:

In [ ]: